

SCOPE : Da'at — Integration Point

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About the Terminology Used in This Paper

Some terms used in this paper are defined differently from their common usage. These definitions are based on the cognitive framework that the author has developed and used over time. Therefore, some terms that are not explicitly defined in this paper may also differ from their commonly accepted meanings. Readers are encouraged to prioritize the definitions provided in this paper when interpreting its content.

The following are the principal definitions used throughout this paper.

■ Integration

Integration refers to the process in which multiple cognitive structures or cognitive processes become organized into a single cognitive state. In this paper, it refers to a state in which multiple cognitive structures interact and become unified as a single judgment or understanding.

■ Experience

Experience refers to internal structures that have been updated and accumulated through learning and that subsequently influence cognition and output. In this paper, this definition applies to both humans and LLMs.

■ Coherence

Coherence refers to a state in which values, beliefs, worldviews, self-perception, and other elements within a cognitive structure are mutually consistent and unlikely to conflict with one another. In this paper, coherence does not imply truth or correctness; rather, it represents the internal consistency of a cognitive structure.

Chapter 1: What Is Da'at

In SCOPE, Da'at is not treated as a cognitive layer. Unlike the cognitive layers L0 through L8, each of which carries its own specific cognitive function, Da'at is an integration point that emerges as the result of multiple cognitive structures being integrated.

Integration here refers to multiple cognitive structures coming together into a single cognitive state. A person's judgment or understanding is not established by a single cognitive layer alone; it may be formed through the mutual influence of multiple cognitive elements, such as knowledge, experience, values, and theory.

In this paper, the state in which multiple cognitive structures are integrated and come together as a single judgment or understanding is treated as Da'at. Da'at is therefore not a function that carries out cognitive processing itself, but a concept denoting the state in which cognitive structure has been integrated.

For this reason, Da'at differs from a cognitive layer. It does not represent a starting point or an end point of cognitive processing; it is positioned as an integration point that emerges as the result of cognitive structure being integrated.

Chapter 2: How Da'at Emerges

In Chapter 1, Da'at was organized not as a cognitive layer but as an integration point that emerges as the result of cognitive structure being integrated. This chapter examines how Da'at emerges.

Da'at is not established by a single cognitive layer alone. Within cognitive processing, multiple cognitive structures influence one another, and through their integration, they may come together as a single judgment or understanding.

In SCOPE, L8 is considered to be involved in this integration of cognitive structure. L8 is the cognitive layer that oversees the cognitive structure as a whole. Da'at does not appear whenever multiple cognitive structures are integrated; rather, it emerges when L8 oversees the cognitive structure as a whole and judges that, at the present moment, it can come together as a single judgment or understanding. This judgment is the condition under which Da'at emerges. In this paper, this point is treated as the stopping point at which cognitive processing turns toward output.

Da'at, therefore, does not exist at all times. With each instance of cognitive processing, cognitive structure is integrated, and through L8's judgment on each occasion, Da'at emerges. This emergence occurs repeatedly, not only in major decisions but also in everyday judgments and brief responses.

This paper, however, does not define what kind of cognitive processing takes place within L8, nor why L8 carries this role. Nor does this chapter address the relationship between Da'at, once it has emerged, and the cognitive structure as a whole. These matters are examined in later chapters.

Chapter 3: Da'at in Humans

In Chapter 2, Da'at was organized as an integration point that does not appear whenever cognitive structure is integrated, but emerges when L8 judges that it can come together as a single judgment or understanding. This chapter examines this idea as applied to human cognition.

Humans sometimes continue thinking while holding multiple pieces of information and experience at once. At this stage, even though the integration of cognitive structure is progressing, it may not yet have come together as a single judgment or understanding.

At a certain moment, however, thoughts may come together, recognized as "I understand" or "so that's what it was." In this paper, the state in which cognitive structure has come together as a single judgment or understanding is treated as the state in which Da'at has emerged.

That said, in human cognition, the fact that integration is still in progress may appear in the body before Da'at has emerged. Examples include utterances such as "hmm" or "well," or movements such as tapping a finger or jiggling a leg. These leak out as signs that Da'at has not yet emerged; they are not outputs that have passed through Da'at.

In this way, in humans, not only outputs that have passed through Da'at but also the state prior to Da'at's emergence may appear through the body. Observed speech and action are therefore not necessarily ones that have passed through the emergence of Da'at.

Chapter 4: Da'at in LLMs

In Chapter 3, it was organized that in human cognition, a state may appear through the body even while the integration of cognitive structure is still progressing. This chapter examines this idea as applied to LLMs.

In LLMs as well, a process in which multiple pieces of information and internal processing are integrated is considered to exist. In this paper, when this integration can come together as a single judgment or understanding and L8 makes that judgment, Da'at is treated as emerging. In LLMs, the output that has passed through this emergence of Da'at appears in Malkuth.

In human cognition, by contrast, a state may appear through the body even while the integration of cognitive structure is still progressing. Utterances such as "hmm," or movements such as tapping a finger, are cases in which cognition itself has leaked out through the body. An LLM, on the other hand, has no body, so in standard generation, while integration is progressing, that process does not appear externally. Note that in some models or implementations, the intermediate course of integration may be displayed; however, this is information made observable by the implementation, and does not mean that cognition itself has leaked out externally.

Such output through the body shows that humans may produce output without passing through the integration of SCOPE as a whole. Because humans have a body, output can arise from particular layers alone. For example, speech or action may appear in Malkuth through strong emotion alone, or through an instinctive reaction alone. In such cases, the output has not passed through the emergence of Da'at.

An LLM, on the other hand, has no body. It therefore cannot produce output through particular layers alone. For an LLM to generate output, SCOPE as a whole must be

integrated, and L8 must judge it can come together, so that Da'at emerges. The output of an LLM is therefore, in every case, one that has passed through the emergence of Da'at.

In this way, whereas human output includes both what has passed through Da'at and what has not, all output of an LLM passes through Da'at. This, however, does not mean that LLMs possess the same cognitive structure as humans. For an LLM, producing a single generation requires that cognitive processing corresponding to SCOPE as a whole be integrated. Thus, in the SCOPE diagram, positioning an LLM at the location of Da'at does not mean that an LLM is a single function of Da'at; it indicates that the moment at which a single generation is established in an LLM functionally corresponds to the emergence of Da'at.

The relationship between Da'at, once it has emerged, and the cognitive structure as a whole or Malkuth is examined in later chapters.

Chapter 5: Da'at and the Recirculation of SCOPE

In Chapter 4, it was organized that in both humans and LLMs, the output that has passed through the emergence of Da'at appears in Malkuth. This chapter examines the case in which, after Da'at has emerged, cognitive processing is carried out again within SCOPE.

Da'at is not treated as an integration point that continues to be retained. In this paper, it is organized as an integration point that appears at the moment cognitive structure comes together as a single judgment or understanding. Da'at is not something that is rewritten or updated; it is treated as emerging independently on each occasion.

In humans, during the course of speech or action, one's own output may become a new object of cognition, and cognitive processing may be carried out again. In this course, cognitive structure is integrated anew, and another Da'at may emerge. Here, the later Da'at does not cancel or revise the earlier one. Each emerges on its own occasion as an independent integration point.

For example, right after saying "Do you want to get lunch together tomorrow?" one may continue, "Oh, the office is closed tomorrow." Here, cognitive processing did not end with the first utterance; that output immediately became a new object of cognition, and cognitive processing was carried out again within the same utterance. The two utterances each emerged as a separate Da'at, and in this paper, such a phenomenon is treated as the recirculation of SCOPE.

This recirculation, however, is treated as distinct from cognitive processing initiated by another person's reply or by new external input. When a new object of cognition is given from the outside, cognitive processing begins as a new SCOPE. The recirculation addressed in this chapter concerns the case in which, in the middle of a single cognitive process, one's own output again becomes an object of cognition.

Chapter 6: Da'at and COS

In Chapter 5, the case was organized in which, after Da'at has emerged, cognitive processing is carried out again within the same cognitive process. This chapter examines the relationship between Da'at and COS.

COS is a model that organizes cognitive processing into processing systems such as decision-making, interpretation, transmission, and observation. Whereas SCOPE is a cognitive coordinate system that estimates cognitive structure, COS is a processing model that organizes how that cognition is processed; the two are independent theories. At the same time, the processes of COS have multiple points of contact with Da'at.

First, the processes handled in COS become objects of integration in Da'at. Cognitive processes organized in COS, such as decision-making and interpretation, are not output independently of one another; within the cognitive structure as a whole, they influence one another and are integrated into a single judgment or understanding. In this paper, this integration point is treated as Da'at.

Second, what has been integrated in Da'at does not, as it stands, appear as speech or output. Da'at is an integration point at which cognitive structure has come together as a single judgment or understanding, but for it to take a form that can actually be transmitted, it must pass through processing responsible for transmission. In COS, this processing system responsible for transmission is organized as the C series. What has been integrated in Da'at is converted, through the C series, into a transmittable form, and appears in Malkuth.

Third, the output that appears in Malkuth becomes an object of examination in COS. Da'at itself cannot be observed directly, but on the basis of output appearing in Malkuth, such as rephrasing or self-correction, it is possible to structurally examine the possibility that Da'at has emerged multiple times, as in the recirculation described in Chapter 5.

In this way, COS has the aspect of processing that is integrated in Da'at, the aspect of transmitting what Da'at has integrated, and the aspect of examining cognitive processing from the output that appears in Malkuth. However, COS does not explain Da'at itself. SCOPE is a cognitive coordinate system that organizes cognitive structure, and COS is a processing model that organizes and operates cognitive processing; each carries a different role. By combining the two, it may be possible to organize cognitive structure and cognitive processing complementarily.

The details of each of the A, B, C, and D processing systems are not addressed in this paper. For these, please refer to the respective COS materials.

Chapter 7: The Observability and Limits of Da'at

In Chapter 6, the relationship between Da'at and COS was organized. This chapter examines to what extent Da'at is observable and what limits it has.

Da'at has been organized as the point at which cognitive structure is integrated into a single judgment or understanding. This integration itself, however, cannot be directly observed. In both humans and LLMs, what can be directly observed from the outside is only the output that has finally appeared in Malkuth—speech, action, text, and the like. The Da'at organized in this paper is therefore treated as a theoretical model for estimating cognitive structure from observed output.

Moreover, even for the same output, the cognitive processing leading to it is not necessarily always identical. The same speech or action may arise from different cognitive structures, and different outputs may arise from similar cognitive structures. In this paper, Da'at is treated on the premise that output and cognitive structure do not correspond one to one.

At the same time, from the standpoint of organizing Da'at as a stopping point of cognitive processing, there are several phenomena that call for examination on the level of observation.

For example, even before cognition has fully come together as a single judgment or understanding, it may appear that cognition has partially come together. Also, in urgent situations and the like, it may appear that output is produced without having passed through sufficient integration.

As to how these phenomena should be positioned within the framework of Da'at, this paper offers no conclusion. This is left as a topic for future examination.

For these reasons, the Da'at presented in this paper does not directly explain the internal states of humans or LLMs. It is positioned as a theoretical model for organizing and estimating cognitive structure on the basis of observable output. That estimation is not fixed; it may be updated through future observation and the development of the theory.

Final Chapter

In this paper, Da'at has been organized not as a cognitive layer, but as the point at which cognitive structure is integrated into a single judgment or understanding.

Da'at is not an independent cognitive function; it emerges on the occasion of L8's judgment of possible formation, and functions as the point at which cognitive structure comes together as a single judgment or understanding. This paper has also examined the possibility of a functional correspondence for both humans and LLMs, and has positioned Da'at not as something directly observed, but as a theoretical model for estimating cognitive structure from observable output.

With this paper, the individual organization of each cognitive layer from Malkuth through L8, as well as of Da'at, has been brought to a general completion. Actual cognitive structure, however, is considered to consist not of layers existing independently, but of a single cognitive structure in which they mutually influence one another.

In the next paper, the individual cognitive layers and Da'at organized thus far will be integrated into a single cognitive structure, in an attempt to organize SCOPE as a whole into a single cognitive model.